

Here are the revised, comprehensive notes for Sections 4.6 through 4.10. I have broken down the difficult concepts and added **concrete numerical examples** for every section to make the math clearer.

4.6 Forward Rates

What are they?

A forward rate is an interest rate applicable to a financial transaction that will take place in the future. It is "implied" by the current spot rates (zero rates) we see today.

The "No-Arbitrage" Logic

Investing for 2 years should yield the exact same result as investing for 1 year, then rolling that money over for a second year at the *forward rate*. If these didn't match, traders could make free money (arbitrage).

Formula (Continuous Compounding)

To find the forward rate (R_F) between year T_1 and year T_2 :

$$R_F = \frac{R_2 T_2 - R_1 T_1}{T_2 - T_1}$$

Example: Calculating a Forward Rate

Imagine the current market rates are:

- **1-Year Zero Rate (R_1):** 3.0%
- **2-Year Zero Rate (R_2):** 4.0%

Question: What is the implied rate for the 2nd year (the 1-year rate starting one year from now)?

$$R_F = \frac{(0.04 \times 2) - (0.03 \times 1)}{2 - 1}$$

$$R_F = \frac{0.08 - 0.03}{1} = \mathbf{5.0\%}$$

Why this makes sense:

- **Option A (Invest for 2 years):** \$100 grows at 4% for 2 years $\rightarrow 100e^{0.04 \times 2} = \108.33
- **Option B (Invest 1 year, then reinvest):** \$100 grows at 3% for 1 year $\rightarrow \$103.05$. Then, that amount grows at **5%** for the next year $\rightarrow 103.05e^{0.05 \times 1} = \108.33 .
- **Result:** Both give \$108.33. The 5% forward rate keeps the market in balance.

4.7 Forward Rate Agreements (FRAs)

What are they?

An FRA is a contract between two parties to "lock in" an interest rate for a future period. It's like a bet on where interest rates will go.

- **Borrower (Fixed-Rate Payer):** Worries rates will go *up*. Wants to pay a fixed rate.
- **Lender (Fixed-Rate Receiver):** Worries rates will go *down*. Wants to receive a fixed rate.

How they are Valued

An FRA is valued by comparing the **Fixed Rate** (R_K) you agreed to vs. the current **Forward Rate** (R_F) in the market.

Valuation Formula

Value to the person receiving the Fixed Rate:

$$V = L \times (R_K - R_F) \times (T_2 - T_1) \times e^{-R_2 T_2}$$

(Where L is principal, and $e^{-R_2 T_2}$ discounts the profit back to today)

Example: Valuing an FRA

- **Scenario:** You entered an FRA to **receive a fixed rate of 4.0%** on \$100 million for a 1-year period, starting 2 years from now.
- **Current Market:** Interest rates have risen. The calculated forward rate for that period is now **5.0%**.
- **Risk-Free Rate** (R_2): 3% (for discounting).

Analysis: You locked in receiving 4%, but the market is now paying 5%. You are losing money on this contract.

$$V = 100,000,000 \times (0.04 - 0.05) \times (1) \times e^{-0.03 \times 3}$$

$$V = 100,000,000 \times (-0.01) \times 0.9139$$

$$V = -\$913,900$$

Result: The contract has a **negative value** of roughly \$913,900 to you.

4.8 Duration

What is it?

Duration measures how sensitive a bond's price is to changes in interest rates.

- Think of it as the "weighted average time" until you get your money back.
- **Zero-coupon bond:** Duration equals its maturity (e.g., a 5-year zero duration is 5).
- **Coupon bond:** Duration is *less* than maturity (because you get some money back early via coupons).

The Key Rule:

$$\text{Percentage Change in Price} \approx -(\text{Duration}) \times (\text{Change in Yield})$$

Example: Using Duration to Predict Price Drops

- **Bond:** A 5-year bond with a Price of \$100.
- **Duration:** Calculated to be **4.5 years**.
- **Scenario:** Interest rates (yields) suddenly **increase by 0.1%** (10 basis points).

Question: How much does the bond price change?

$$\Delta P \approx -4.5 \times (+0.001)$$

$$\Delta P \approx -0.0045 \text{ or } -0.45\%$$

Result: The bond price will drop by approximately **0.45%** (falling from \$100 to \$99.55).

- *Note the negative sign:* When rates go UP, prices go DOWN.

4.9 Convexity

What is it?

Duration is a straight-line estimate (linear). But the real relationship between price and yield is curved (convex).

- **The Problem:** For *large* rate changes, Duration isn't accurate enough. It is too pessimistic (it predicts prices will drop more than they actually do when rates rise).
- **The Fix:** We add "Convexity" to account for the curvature.

Analogy

- **Duration** is like using a straight ruler to measure a curved hill. It works for a few steps, but eventually, the gap gets too big.
- **Convexity** bends the ruler to match the hill.

Example: Why Convexity Matters

Using the previous example (Duration 4.5 years), suppose rates rise by a massive **2.0%**.

- **Duration Prediction:** Price drops by $4.5 \times 2\% = 9\%$.
- **Actual Reality (due to convexity):** The price might actually drop only 8.2%.
- **Conclusion:** The bond performed *better* than the duration formula predicted. A bond with **higher convexity** is generally better because it drops less when rates rise and rises more when rates fall.

4.10 Theories of the Term Structure

Question: Why are 10-year rates usually higher than 1-year rates? (Why is the curve upward sloping?)

1. Expectations Theory

- *Concept:* Rates are higher simply because the market *expects* rates to go up in the future.
- *Example:* If the 1-year rate is 2% and the 2-year rate is 4%, it's because the market thinks next year's rate will be roughly 6%. It's purely math.

2. Market Segmentation Theory

- *Concept:* Different investors live in different "buckets."
- *Example:* Pension funds buy 30-year bonds. Day traders buy 1-year bills. They don't talk to each other. Supply and demand in the "30-year bucket" sets the 30-year rate, regardless of what the "1-year bucket" is doing.

3. Liquidity Preference Theory (The most popular one)

- *Concept:* Cash is king (Liquidity).
- *Logic:*

- **Investors** hate locking up money for 10 years. They demand a "bonus" (premium) to do it.
- **Borrowers** (like homeowners) love locking in rates for 30 years for safety. They are willing to pay extra for that safety.
- *Result:* Long-term rates are naturally higher than short-term rates because of this "Liquidity Premium." This explains why the yield curve is usually upward sloping, even if we don't expect rates to change much.